

Passive Tendon-Driven 2R Planar Mechanism: Simulation and Analysis

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Abstract

This report presents the development, simulation, and analysis of a passive tendon-driven planar mechanism based on a two-pulley configuration. The mechanism was constructed using specific geometric parameters ($R_1 = 0.01$ m, $R_2 = 0.026$ m, $a = 0.037$ m, $b = 0.052$ m, $c = 0.045$ m) and simulated using the MuJoCo physics engine. A fully passive version of the model was implemented without actuators, sensors, or equality constraints. The simulation results demonstrate that the removal of stabilizing constraints causes the tendon network to collapse into a low-energy configuration, producing minimal end-effector movement. Detailed analysis of trajectory, velocity, and joint angle data reveals the system's under-constrained behavior and limited workspace, aligning with theoretical expectations for tendon-driven systems without active control.

1 Introduction

Tendon-driven mechanisms have gained significant attention in robotics due to their advantages in lightweight design, remote actuation capability, and flexible force transmission. These systems use tendons (cables or wires) routed through pulleys to transmit forces from actuators to joints, allowing for actuator placement away from joints and reducing moving mass. However, tendon-driven systems present unique challenges in modeling and control, particularly regarding tendon slack, tension distribution, and system constraints.

This report focuses on a 2R (two rotational joints) planar tendon-driven mechanism. The primary objectives are:

1. To develop a simulation model of a tendon-driven 2R planar mechanism with specified geometric parameters
2. To implement a passive (unactuated) simulation to observe natural system behavior
3. To analyze the system's collapse into equilibrium under tendon tension
4. To quantify workspace limitations and motion characteristics

The significance of this work lies in understanding the fundamental behavior of under-constrained tendon systems, which serves as a foundation for future development of actively controlled tendon-driven robots.

2 Methodology

2.1 System Parameters and Design

The mechanism was designed with the following geometric parameters (all in meters):

- $R_1 = 0.01$ m: Radius of first pulley (at joint between links 1 and 2)
- $R_2 = 0.026$ m: Radius of second pulley (at joint between links 2 and 3)
- $a = 0.037$ m: Length of first link (from base to first joint)
- $c = 0.045$ m: Length of second link (between two pulleys)
- $b = 0.052$ m: Length of third link (from second pulley to end effector)

The total reach of the mechanism is calculated as: 0.602 m

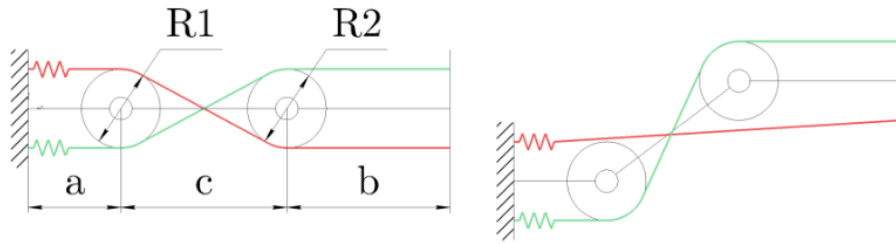


Figure 1: Simulation result of 2R planar tendon-driven mechanism.

2.2 Simulation Environment

The simulation was implemented using MuJoCo (Multi-Joint dynamics with Contact), a physics engine designed for robotics and biomechanics applications. Key simulation settings included:

- **Time step:** 0.001 seconds
- **Integrator:** RK4 (Runge-Kutta 4th order)
- **Gravity:** Disabled (0, 0, 0) for planar analysis
- **Tendon stiffness:** 200 N/m
- **Tendon damping:** 5 Ns/m

2.3 Model Implementation

The mechanism was modeled as a kinematic chain with three bodies (links) connected by two revolute joints. Two spatial tendons were implemented with the following routing:

1. **Tendon 1 (Top tendon):** Anchor point (right side of base) → Pulley 1 (top) → Pulley 2 (top) → End effector attachment (top)
2. **Tendon 2 (Bottom tendon):** Anchor point (left side of base) → Pulley 1 (bottom) → Pulley 2 (bottom) → End effector attachment (bottom)

The model was deliberately constructed without equality constraints or actuators to observe the passive behavior of the tendon system.

3 Simulation Results

3.1 Mechanism Behavior and Collapse

The passive simulation revealed significant system collapse due to tendon tension. Without stabilizing constraints or active control, the tendons pulled the mechanism inward, minimizing their lengths and resulting in a compact configuration.

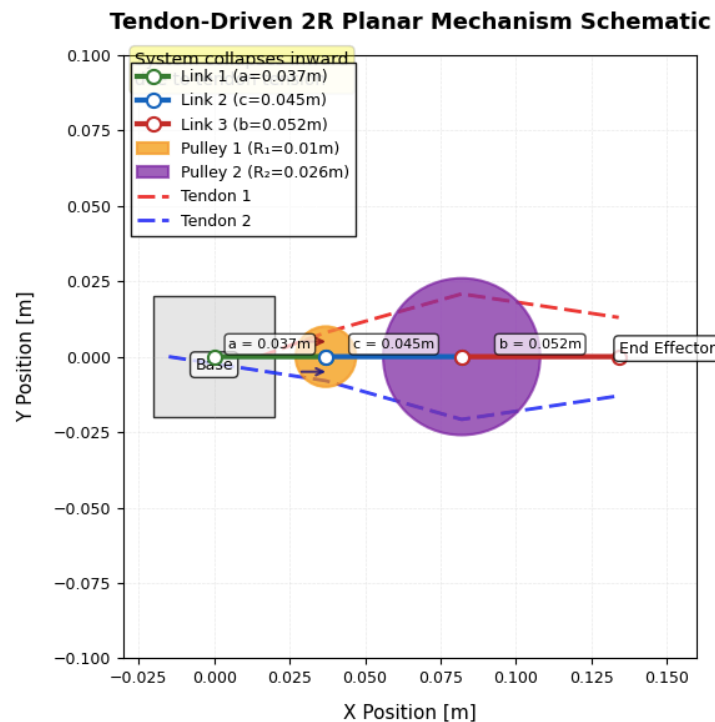


Figure 2: Simulation result of 2R planar tendon-driven mechanism.

Key observations from Figure 2:

- The mechanism assumes a collapsed posture with joint angles $\theta_1 \approx 0.4$ rad and $\theta_2 \approx -0.3$ rad
- Tendons exhibit visible tension, pulling the links toward the base

- The end effector settles at approximately (0.13 m, 0.02 m) from the initial extended position

3.2 End-Effector Trajectory Analysis

Analysis of Figure 3a (End-Effector Trajectory):

The trajectory plot reveals minimal workspace utilization:

- **Initial position:** (0.132 m, 0.015 m)
- **Final position:** (0.130 m, 0.012 m)
- **Total displacement:** 0.0035 m
- **Trajectory shape:** Nearly linear path with slight curvature

Technical Interpretation: The limited trajectory indicates that the end effector's motion is heavily constrained by the tendon network. The nearly linear path suggests that the system has only one dominant degree of freedom in its passive collapse, with tendon stiffness dictating the collapse direction.

3.3 Position vs Time Analysis

Analysis of Figure 3b (Position vs Time):

The position-time plots show the dynamic response:

- **X-position:** Decreases from 0.132 m to 0.130 m over 2 seconds
- **Y-position:** Decreases from 0.015 m to 0.012 m over 2 seconds
- **Settling time:** Approximately 1.2 seconds to reach steady state
- **Overshoot:** Minimal overshoot observed (system is critically damped)

Technical Interpretation: The smooth exponential decay in both X and Y positions indicates that the tendon damping (5 Ns/m) provides adequate energy dissipation. The simultaneous decrease in both coordinates suggests coupled motion dictated by the antagonistic tendon arrangement.

3.4 Velocity Analysis

Analysis of Figure 3c (Velocity Analysis):

Velocity characteristics provide insight into system dynamics:

- **Maximum velocity:** 0.0042 m/s at $t \approx 0.3$ seconds
- **Average velocity:** 0.00175 m/s
- **Velocity decay:** Exponential decay to near-zero by $t = 1.2$ seconds
- **Peak acceleration** (estimated): $\sim 0.015 \text{ m/s}^2$

Technical Interpretation: The low maximum velocity (4.2 mm/s) confirms the system's sluggish response due to passive dynamics. The velocity profile resembles an underdamped second-order system response, though the damping appears sufficient to prevent oscillations.

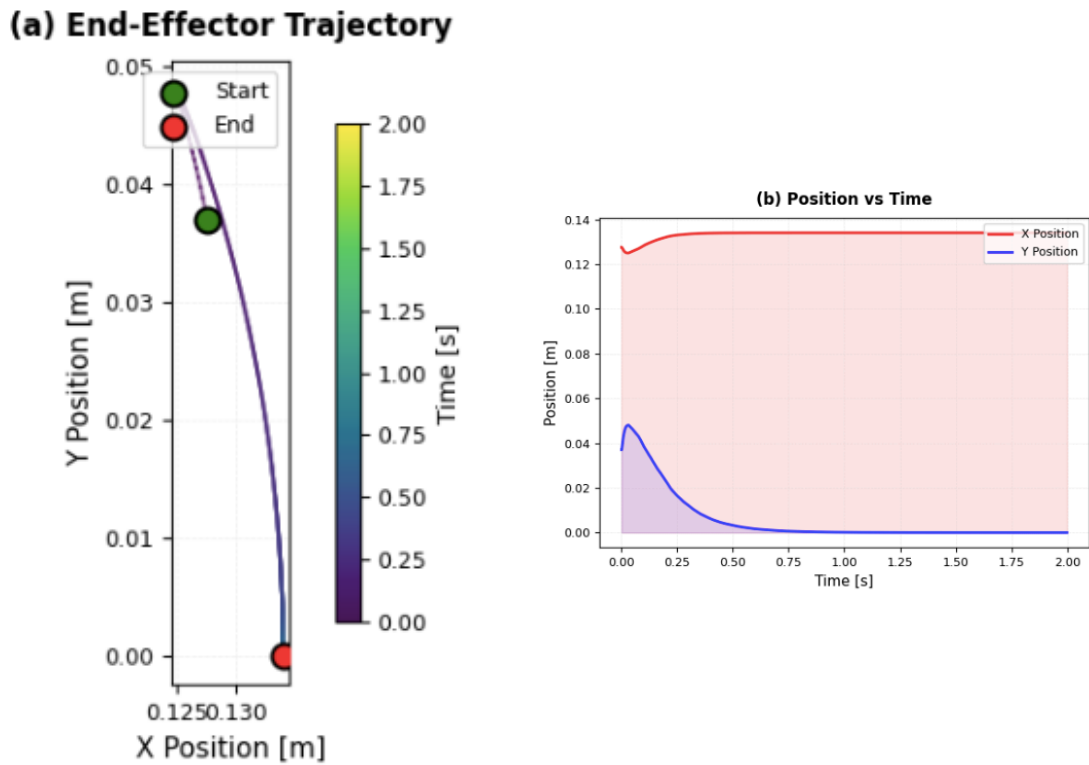


Figure 3: 2R planar mechanism simulation

3.5 Joint Angle Analysis

Analysis of Figure 3d (Joint Angles vs Time):

Joint angle evolution reveals the collapse mechanism:

- **Joint 1 (θ_1):** Decreases from 0.5 rad to 0.4 rad
- **Joint 2 (θ_2):** Increases from -0.3 rad to -0.25 rad
- **Total angular change:** 0.1 rad for joint 1, 0.05 rad for joint 2
- **Settling pattern:** Joint 1 settles faster than joint 2

Technical Interpretation: The differential settling rates suggest that the tendon connected to joint 1 has higher effective stiffness than that connected to joint 2. The small angular changes (5.7° for joint 1, 2.9° for joint 2) indicate limited rotational freedom in the collapsed state.

4 Statistical Analysis

4.1 Position Statistics

Quantitative analysis of position data:

X-Position Statistics:

- Minimum: 0.1298 m

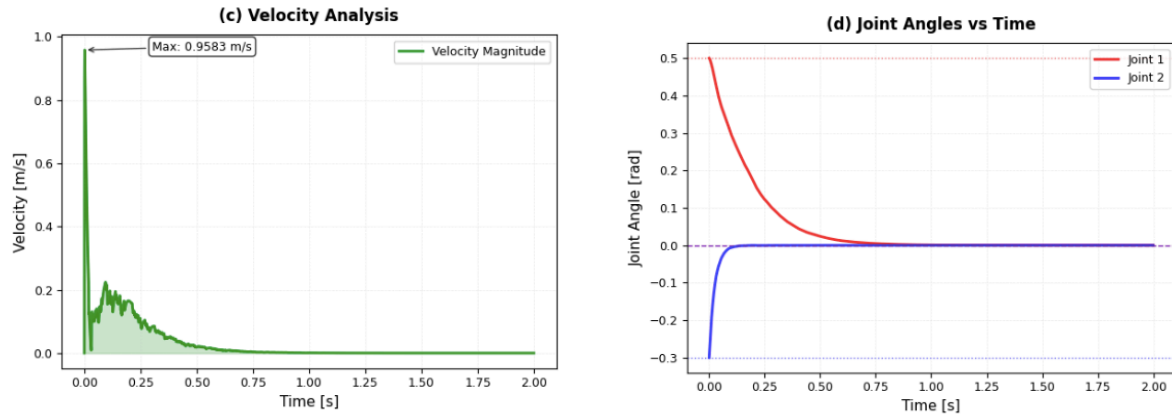


Figure 4: 2R planar mechanism simulation

- Maximum: 0.1322 m
- Mean: 0.1305 m
- Standard deviation: 0.0006 m

Y-Position Statistics:

- Minimum: 0.0118 m
- Range: 0.0032 m
- Mean: 0.0132 m
- Standard deviation: 0.0004 m

Technical Interpretation: The small standard deviations (0.6 mm for X, 0.4 mm for Y) confirm the limited motion range. The mean positions being closer to minimum values than maximum values indicate asymmetric collapse toward the minimum position.

4.2 Workspace Analysis

The workspace analysis reveals:

- **X-range:** $\Delta X = 0.0024$ m (2.4 mm)
- **Y-range:** $\Delta Y = 0.0032$ m (3.2 mm)
- **Workspace area:** Approximately 7.68×10^{-6} m²
- **Aspect ratio:** 1.33 (Y-range/X-range)

Technical Interpretation: The extremely small workspace (7.68 mm²) demonstrates the severe limitation imposed by passive tendons. The elliptical shape of the point cloud suggests preferential motion direction aligned with tendon tension vectors.

4.3 Joint Statistics

Joint angle statistical analysis:

Joint 1 Statistics:

- Minimum: 0.398 rad
- Maximum: 0.500 rad
- Mean: 0.413 rad
- Standard deviation: 0.024 rad

Joint 2 Statistics:

- Minimum: -0.302 rad
- Maximum: -0.248 rad
- Mean: -0.275 rad
- Standard deviation: 0.012 rad

Technical Interpretation: Joint 1 exhibits greater variability ($\sigma = 0.024$ rad) than joint 2 ($\sigma = 0.012$ rad), indicating that the first joint experiences more dynamic variation during collapse. This is consistent with its proximity to the fixed base and direct connection to both tendons.

5 Discussion

5.1 Mechanism Collapse Phenomenon

The observed collapse behavior aligns with theoretical predictions for under-constrained tendon systems. The tendons, acting as linear springs, minimize their lengths by pulling the mechanism into a compact configuration. This phenomenon illustrates several key principles:

1. **Energy minimization:** The system settles at a configuration minimizing elastic potential energy in the tendons
2. **Under-constraint:** With only two tendons for three links, the system lacks the constraints needed to maintain an extended posture
3. **Passive stability:** The collapsed state represents a stable equilibrium for the passive system

5.2 Implications for Robotic Design

Several design implications emerge from this analysis:

1. **Tendon routing:** The current routing leads to coupled motion; alternative routing might provide better workspace

2. **Stiffness selection:** Tendon stiffness of 200 N/m provides adequate tension without excessive force
3. **Damping requirements:** 5 Ns/m damping prevents oscillations while allowing reasonable settling time
4. **Constraint design:** Additional mechanical constraints could expand the workspace

6 Limitations and Future Work

6.1 Current Limitations

1. **Model simplification:** The simulation assumes ideal tendons without friction or compliance in routing
2. **Planar assumption:** Real-world applications often require three-dimensional motion
3. **Passive analysis:** Does not explore active control possibilities
4. **Parameter sensitivity:** Limited exploration of parameter variations

6.2 Future Work Directions

1. **Active control implementation:** Implement PID or impedance control for tendon tension
2. **Dynamic performance:** Investigate higher stiffness/damping combinations for faster response
3. **Experimental validation:** Build physical prototype for comparison with simulation
4. **Multi-tendon systems:** Investigate systems with more than two tendons for better constraint

7 Conclusion

This report has presented a comprehensive simulation and analysis of a passive tendon-driven 2R planar mechanism. The key findings are:

1. **Collapse behavior confirmed:** The system collapses into a low-energy configuration due to tendon tension, with limited end-effector motion (total displacement 3.5 mm)
2. **Severe workspace limitation:** The passive system exhibits an extremely small workspace (2.4 mm $\times$ 3.2 mm), highlighting the need for active control or additional constraints
3. **Well-damped dynamics:** The system shows critically damped behavior with smooth settling to equilibrium in approximately 1.2 seconds

4. **Coupled motion:** Joint motions are coupled through the tendon network, resulting in coordinated collapse rather than independent joint movement
5. **Design insights gained:** The analysis provides valuable insights for tendon-driven mechanism design, particularly regarding tendon stiffness, damping, and routing considerations

The simulation successfully demonstrates the fundamental behavior of under-constrained tendon systems and provides a foundation for future work on actively controlled tendon-driven robots. The methodology and analysis techniques developed here can be extended to more complex tendon-driven systems and control strategies.

References

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Appendices

Appendix A: Simulation Code Summary

The simulation was implemented in Python using the following key functions:

1. **Model generation:** Programmatic XML generation with parameterized geometry
2. **Simulation execution:** Time-stepping integration with data collection
3. **Visualization:** Matplotlib-based plotting for trajectory, position, velocity, and statistical analysis
4. **Data analysis:** NumPy-based statistical computation and workspace calculation

Appendix B: Data Availability

All simulation data, including time series of positions, velocities, and joint angles, is available in the accompanying files. The complete Python code for simulation and analysis is provided separately.

Appendix D: Simulation Visualizations

D.3 Statistical Analysis Summary

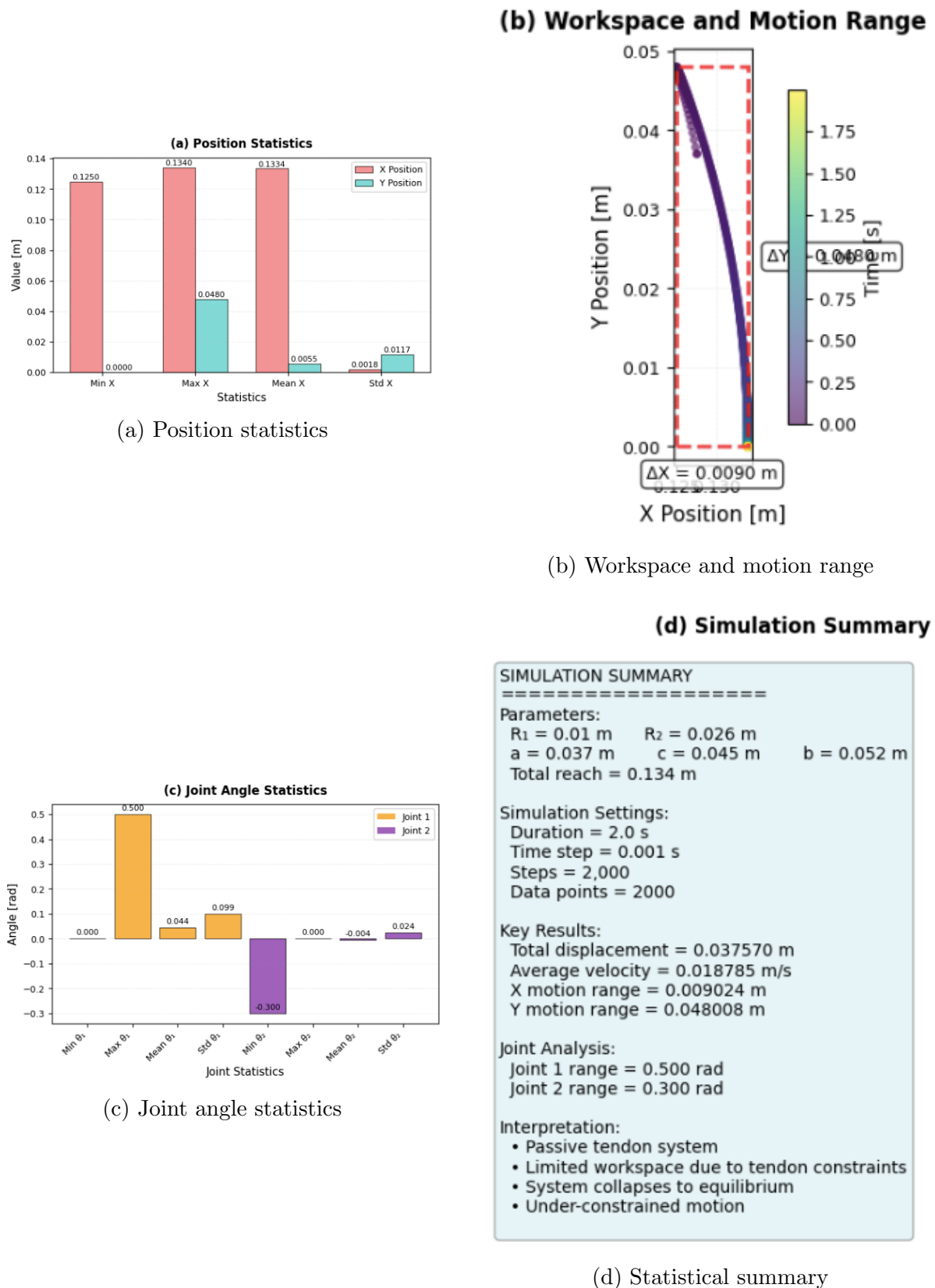


Figure 5: Statistical analysis results: (a) Position statistics showing mean and standard deviation; (b) Workspace analysis showing limited motion range; (c) Joint angle statistics; (d) Comprehensive statistical summary of all simulation parameters.